

Exercise Sheet for Week 5 - Type inference and polymorphism

Assigned: Monday 28 October
Suggested Deadline: Monday 4 November

Question 1: Recap: type-checking

Give derivations to show that each of the following is a valid typing judgement (using Curry-style terms).

- (a) $\vdash \lambda x. x (\lambda y. x (\lambda z. y)) : ((\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}) \rightarrow \mathbb{B}$

Write $\Gamma = x : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}$ to save space.

$$\frac{\frac{\frac{\frac{\Gamma \vdash x : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}}{\Gamma, y : \mathbb{B} \vdash x : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}} \text{VAR}}{\Gamma, y : \mathbb{B} \vdash x (\lambda z. y) : \mathbb{B}} \text{APP}}{\Gamma, y : \mathbb{B} \vdash x (\lambda y. x (\lambda z. y)) : \mathbb{B}} \text{ABS}}{\vdash \lambda x. x (\lambda y. x (\lambda z. y)) : ((\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}) \rightarrow \mathbb{B}} \text{ABS}$$

- (b) $\vdash \lambda x. (\lambda y. y)x : (\alpha \rightarrow \beta) \rightarrow \alpha \rightarrow \beta$

$$\frac{\frac{\frac{x : \alpha \rightarrow \beta, y : \alpha \rightarrow \beta \vdash y : \alpha \rightarrow \beta}{x : \alpha \rightarrow \beta \vdash \lambda y. y : (\alpha \rightarrow \beta) \rightarrow \alpha \rightarrow \beta} \text{VAR}}{x : \alpha \rightarrow \beta \vdash (\lambda y. y)x : \alpha \rightarrow \beta} \text{APP}}{\vdash \lambda x. (\lambda y. y)x : (\alpha \rightarrow \beta) \rightarrow \alpha \rightarrow \beta} \text{ABS}$$

- (c) $\vdash \lambda f. \text{if } f \text{ true then } f \text{ else } \lambda y. f (f y) : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}$

$$\frac{\frac{\frac{f : \mathbb{B} \rightarrow \mathbb{B} \vdash f : \mathbb{B} \rightarrow \mathbb{B}}{f : \mathbb{B} \rightarrow \mathbb{B} \vdash f \text{ true} : \mathbb{B}} \text{VAR}}{f : \mathbb{B} \rightarrow \mathbb{B} \vdash \text{if } f \text{ true then } f \text{ else } \lambda y. f (f y) : \mathbb{B} \rightarrow \mathbb{B}} \text{APP}}{\vdash \lambda f. \text{if } f \text{ true then } f \text{ else } \lambda y. f (f y) : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}} \text{ITE}$$

where Π_1 is

$$\frac{\frac{\frac{f : \mathbb{B} \rightarrow \mathbb{B}, y : \mathbb{B} \vdash f : \mathbb{B} \rightarrow \mathbb{B}}{\text{VAR}} \quad \frac{\frac{f : \mathbb{B} \rightarrow \mathbb{B}, y : \mathbb{B} \vdash f : \mathbb{B} \rightarrow \mathbb{B}}{\text{VAR}} \quad \frac{f : \mathbb{B} \rightarrow \mathbb{B}, y : \mathbb{B} \vdash y : \mathbb{B}}{\text{VAR}}}{\frac{f : \mathbb{B} \rightarrow \mathbb{B}, y : \mathbb{B} \vdash f y : \mathbb{B}}{\text{APP}}} \text{VAR}$$

$$\frac{f : \mathbb{B} \rightarrow \mathbb{B}, y : \mathbb{B} \vdash f(f y) : \mathbb{B}}{f : \mathbb{B} \rightarrow \mathbb{B} \vdash \lambda y. f(f y) : \mathbb{B} \rightarrow \mathbb{B}} \text{ABS}$$

Question 2: Type-inference, or finding principal types

For each of the following closed Curry-style λ -terms, find its principal type.

For the solutions to this question, we have presented the execution of Wand's algorithm by following the hypothetical typing derivation and labelling the points at which constraints are generated, because this is likely to be clearer than the linear format used in lectures. For the Robinson's algorithm part, it is more important to follow the format used in lectures of having a set of constraints and a substitution which you act on over discrete iterations of the main loop. We have presented a model for how you might set out your working with a couple of minor variations in presentations.

(a) $\lambda x. x (\lambda y. x (\lambda z. y))$

$$\frac{\frac{\frac{\frac{\frac{\frac{\frac{x : \alpha_1, y : \alpha_4 \vdash x : \alpha_6 \rightarrow \alpha_5}{\text{VAR } [4]} \quad \frac{\frac{x : \alpha_1, y : \alpha_4, z : \alpha_7 \vdash y : \alpha_8}{\text{VAR } [6]} \quad \frac{x : \alpha_1, y : \alpha_4 \vdash \lambda z. y : \alpha_6}{\text{ABS } [5]}}{\text{APP}}} \quad \frac{x : \alpha_1, y : \alpha_4 \vdash x (\lambda z. y) : \alpha_5}{\text{ABS } [3]}}{\text{APP}}} \quad \frac{x : \alpha_1 \vdash x (\lambda y. x (\lambda z. y)) : \alpha_2}{\text{VAR } [2]}}{\text{APP}}} \quad \frac{x : \alpha_1 \vdash x (\lambda y. x (\lambda z. y)) : \alpha_2}{\text{ABS } [1]}} \vdash \lambda x. x (\lambda y. x (\lambda z. y)) : \alpha_0$$

where the following constraints are generated:

$$\begin{array}{l} [1] : \alpha_0 = \alpha_1 \rightarrow \alpha_2 \\ [2] : \alpha_1 = \alpha_3 \rightarrow \alpha_2 \\ [3] : \alpha_3 = \alpha_4 \rightarrow \alpha_5 \\ [4] : \alpha_1 = \alpha_6 \rightarrow \alpha_5 \\ [5] : \alpha_6 = \alpha_7 \rightarrow \alpha_8 \\ [6] : \alpha_4 = \alpha_8 \end{array}$$

which we unify as follows (at each stage we underline the constraint being focused on):

$\alpha_0 = \alpha_1 \rightarrow \alpha_2$	$\alpha_0 \mapsto \alpha_1 \rightarrow \alpha_2$	$\alpha_0 \mapsto (\alpha_3 \rightarrow \alpha_2) \rightarrow \alpha_2$	$\alpha_0 \mapsto ((\alpha_4 \rightarrow \alpha_5) \rightarrow \alpha_2) \rightarrow \alpha_2$
$\alpha_1 = \alpha_3 \rightarrow \alpha_2$	$\alpha_1 \mapsto \alpha_3 \rightarrow \alpha_2$	$\alpha_1 \mapsto \alpha_3 \rightarrow \alpha_2$	$\alpha_1 \mapsto (\alpha_4 \rightarrow \alpha_5) \rightarrow \alpha_2$
$\alpha_3 = \alpha_4 \rightarrow \alpha_5$	$\alpha_3 = \alpha_4 \rightarrow \alpha_5$	$\alpha_3 \mapsto \alpha_4 \rightarrow \alpha_5$	$\alpha_3 \mapsto \alpha_4 \rightarrow \alpha_5$
$\alpha_1 = \alpha_6 \rightarrow \alpha_5$	$\alpha_1 \mapsto \alpha_6 \rightarrow \alpha_5$	$\alpha_3 \rightarrow \alpha_2 = \alpha_6 \rightarrow \alpha_5$	$(\alpha_4 \rightarrow \alpha_5) \rightarrow \alpha_2 = \alpha_6 \rightarrow \alpha_5$
$\alpha_6 = \alpha_7 \rightarrow \alpha_8$	$\alpha_6 = \alpha_7 \rightarrow \alpha_8$	$\alpha_6 = \alpha_7 \rightarrow \alpha_8$	$\alpha_6 = \alpha_7 \rightarrow \alpha_8$
$\alpha_4 = \alpha_8$	$\alpha_4 = \alpha_8$	$\alpha_4 = \alpha_8$	$\alpha_4 = \alpha_8$

$$\begin{array}{l}
\alpha_0 \mapsto ((\alpha_4 \rightarrow \alpha_5) \rightarrow \alpha_2) \rightarrow \alpha_2 \\
\alpha_1 \mapsto (\alpha_4 \rightarrow \alpha_5) \rightarrow \alpha_2 \\
\alpha_3 \mapsto \alpha_4 \rightarrow \alpha_5 \\
\hline \alpha_4 \rightarrow \alpha_5 = \alpha_6 \\
\alpha_2 = \alpha_5 \\
\alpha_6 = \alpha_7 \rightarrow \alpha_8 \\
\alpha_4 = \alpha_8
\end{array}$$

$$\begin{array}{l}
\alpha_0 \mapsto ((\alpha_4 \rightarrow \alpha_5) \rightarrow \alpha_2) \rightarrow \alpha_2 \\
\alpha_1 \mapsto (\alpha_4 \rightarrow \alpha_5) \rightarrow \alpha_2 \\
\alpha_3 \mapsto \alpha_4 \rightarrow \alpha_5 \\
\alpha_6 \mapsto \alpha_4 \rightarrow \alpha_5 \\
\hline \alpha_2 = \alpha_5 \\
\alpha_4 \rightarrow \alpha_5 = \alpha_7 \rightarrow \alpha_8 \\
\alpha_4 = \alpha_8
\end{array}$$

$$\begin{array}{l}
\alpha_0 \mapsto ((\alpha_4 \rightarrow \alpha_5) \rightarrow \alpha_5) \rightarrow \alpha_5 \\
\alpha_1 \mapsto (\alpha_4 \rightarrow \alpha_5) \rightarrow \alpha_5 \\
\alpha_3 \mapsto \alpha_4 \rightarrow \alpha_5 \\
\alpha_6 \mapsto \alpha_4 \rightarrow \alpha_5 \\
\hline \alpha_2 \mapsto \alpha_5 \\
\alpha_4 \rightarrow \alpha_5 = \alpha_7 \rightarrow \alpha_8 \\
\alpha_4 = \alpha_8
\end{array}$$

$$\begin{array}{l}
\alpha_0 \mapsto ((\alpha_4 \rightarrow \alpha_5) \rightarrow \alpha_5) \rightarrow \alpha_5 \\
\alpha_1 \mapsto (\alpha_4 \rightarrow \alpha_5) \rightarrow \alpha_5 \\
\alpha_3 \mapsto \alpha_4 \rightarrow \alpha_5 \\
\alpha_6 \mapsto \alpha_4 \rightarrow \alpha_5 \\
\alpha_2 \mapsto \alpha_5 \\
\hline \alpha_4 = \alpha_7 \\
\alpha_5 = \alpha_8 \\
\alpha_4 = \alpha_8
\end{array}$$

$$\begin{array}{l}
\alpha_0 \mapsto ((\alpha_7 \rightarrow \alpha_5) \rightarrow \alpha_5) \rightarrow \alpha_5 \\
\alpha_1 \mapsto (\alpha_7 \rightarrow \alpha_5) \rightarrow \alpha_5 \\
\alpha_3 \mapsto \alpha_7 \rightarrow \alpha_5 \\
\alpha_6 \mapsto \alpha_7 \rightarrow \alpha_5 \\
\alpha_2 \mapsto \alpha_5 \\
\hline \alpha_4 \mapsto \alpha_7 \\
\alpha_5 = \alpha_8 \\
\alpha_7 = \alpha_8
\end{array}$$

$$\begin{array}{l}
\alpha_0 \mapsto ((\alpha_7 \rightarrow \alpha_8) \rightarrow \alpha_8) \rightarrow \alpha_8 \\
\alpha_1 \mapsto (\alpha_7 \rightarrow \alpha_8) \rightarrow \alpha_8 \\
\alpha_3 \mapsto \alpha_7 \rightarrow \alpha_8 \\
\alpha_6 \mapsto \alpha_7 \rightarrow \alpha_8 \\
\alpha_2 \mapsto \alpha_8 \\
\hline \alpha_4 \mapsto \alpha_7 \\
\alpha_5 \mapsto \alpha_8 \\
\alpha_7 = \alpha_8
\end{array}$$

$$\begin{array}{l}
\alpha_0 \mapsto ((\alpha_8 \rightarrow \alpha_8) \rightarrow \alpha_8) \rightarrow \alpha_8 \\
\alpha_1 \mapsto (\alpha_8 \rightarrow \alpha_8) \rightarrow \alpha_8 \\
\alpha_3 \mapsto \alpha_8 \rightarrow \alpha_8 \\
\alpha_6 \mapsto \alpha_8 \rightarrow \alpha_8 \\
\alpha_2 \mapsto \alpha_8 \\
\alpha_4 \mapsto \alpha_8 \\
\alpha_5 \mapsto \alpha_8 \\
\alpha_7 \mapsto \alpha_8
\end{array}$$

Hence, the principal type is $((\alpha \rightarrow \alpha) \rightarrow \alpha) \rightarrow \alpha$.

(b) $\lambda x. (\lambda y. y)x$

$$\frac{\frac{\frac{}{x : \alpha_1, y : \alpha_4 \vdash y : \alpha_5} \text{VAR} \boxed{3}}{x : \alpha_1 \vdash (\lambda y. y) : \alpha_3 \rightarrow \alpha_2} \text{ABS} \boxed{2} \quad \frac{}{x : \alpha_1 \vdash x : \alpha_3} \text{VAR} \boxed{4}}{\frac{x : \alpha_1 \vdash (\lambda y. y)x : \alpha_2}{\vdash \lambda x. (\lambda y. y)x : \alpha_0} \text{ABS} \boxed{1}} \text{APP}$$

where the following constraints are generated:

$$\begin{array}{l}
\boxed{1} : \alpha_0 = \alpha_1 \rightarrow \alpha_2 \\
\boxed{2} : \alpha_3 \rightarrow \alpha_2 = \alpha_4 \rightarrow \alpha_5 \\
\boxed{3} : \alpha_4 = \alpha_5 \\
\boxed{4} : \alpha_1 = \alpha_3
\end{array}$$

Now we unify (here we point to the constraint being used to get to the next state, and use the horizontal bar

to separate the substitution generated so far from the remaining constraints):

$$\begin{array}{c}
\begin{array}{l}
\text{⊢} \alpha_0 = \alpha_1 \rightarrow \alpha_2 \\
\alpha_3 \rightarrow \alpha_2 = \alpha_4 \rightarrow \alpha_5 \\
\alpha_4 = \alpha_5 \\
\alpha_1 = \alpha_3
\end{array}
\quad
\frac{\alpha_0 \mapsto \alpha_1 \rightarrow \alpha_2}{\alpha_3 \rightarrow \alpha_2 = \alpha_4 \rightarrow \alpha_5}
\quad
\frac{\alpha_0 \mapsto \alpha_3 \rightarrow \alpha_2}{\alpha_1 \mapsto \alpha_3}
\quad
\frac{\alpha_0 \mapsto \alpha_3 \rightarrow \alpha_2}{\alpha_1 \mapsto \alpha_3}
\end{array}$$

$$\begin{array}{c}
\begin{array}{l}
\alpha_0 \mapsto \alpha_4 \rightarrow \alpha_2 \\
\alpha_1 \mapsto \alpha_4 \\
\alpha_3 \mapsto \alpha_4
\end{array}
\quad
\begin{array}{l}
\alpha_0 \mapsto \alpha_4 \rightarrow \alpha_5 \\
\alpha_1 \mapsto \alpha_4 \\
\alpha_3 \mapsto \alpha_4
\end{array}
\quad
\begin{array}{l}
\alpha_0 \mapsto \alpha_5 \rightarrow \alpha_5 \\
\alpha_1 \mapsto \alpha_5 \\
\alpha_3 \mapsto \alpha_5
\end{array}
\end{array}$$

$$\begin{array}{c}
\text{⊢} \alpha_2 = \alpha_5 \\
\alpha_4 = \alpha_5
\end{array}
\quad
\begin{array}{c}
\alpha_2 \mapsto \alpha_5 \\
\text{⊢} \alpha_4 = \alpha_5
\end{array}
\quad
\begin{array}{c}
\alpha_2 \mapsto \alpha_5 \\
\alpha_4 \mapsto \alpha_5
\end{array}$$

Hence, the principal type is $\alpha \rightarrow \alpha$.

(c) $\lambda f. \text{if } f \text{ true then } f \text{ else } \lambda y. f(fy)$

$$\frac{\frac{\frac{f : \alpha_1 \vdash f : \alpha_3 \rightarrow \mathbb{B}}{\text{VAR } [2]} \quad \frac{f : \alpha_1 \vdash \text{true} : \alpha_3}{\text{T } [3]}}{\frac{f : \alpha_1 \vdash f \text{ true} : \mathbb{B}}{\text{APP}}} \quad \frac{f : \alpha_1 \vdash f : \alpha_2}{\text{VAR } [4]} \quad \Pi_1}{\frac{f : \alpha_1 \vdash \text{if } f \text{ true then } f \text{ else } \lambda y. f(fy) : \alpha_2}{\vdash \lambda f. \text{if } f \text{ true then } f \text{ else } \lambda y. f(fy) : \alpha_0} \text{ITE}}{\text{ABS } [1]}$$

where Π_1 is

$$\frac{\frac{\frac{f : \alpha_1, y : \alpha_4 \vdash f : \alpha_6 \rightarrow \alpha_5}{\text{VAR } [6]} \quad \frac{\frac{f : \alpha_1, y : \alpha_4 \vdash f : \alpha_7 \rightarrow \alpha_6}{\text{VAR } [7]} \quad \frac{f : \alpha_1, y : \alpha_4 \vdash y : \alpha_7}{\text{VAR } [8]}}{\frac{f : \alpha_1, y : \alpha_4 \vdash fy : \alpha_6}{\text{APP}}} \quad \frac{f : \alpha_1, y : \alpha_4 \vdash f(fy) : \alpha_5}{\text{ABS } [5]}}{\frac{f : \alpha_1 \vdash \lambda y. f(fy) : \alpha_2}{\text{ABS } [5]}}$$

where the following constraints are generated:

$$\begin{array}{l}
[1] : \alpha_0 = \alpha_1 \rightarrow \alpha_2 \\
[2] : \alpha_1 = \alpha_3 \rightarrow \mathbb{B} \\
[3] : \alpha_3 = \mathbb{B} \\
[4] : \alpha_1 = \alpha_2 \\
[5] : \alpha_2 = \alpha_4 \rightarrow \alpha_5 \\
[6] : \alpha_1 = \alpha_6 \rightarrow \alpha_5 \\
[7] : \alpha_1 = \alpha_7 \rightarrow \alpha_6 \\
[8] : \alpha_4 = \alpha_7
\end{array}$$

Since we are only interested in finding the principal type, we only need to record what the substitution does

to α_0 as we unify:

$$\begin{array}{llll}
\alpha_0 = \alpha_1 \rightarrow \alpha_2 & \alpha_0 = \alpha_1 \rightarrow \alpha_2 & \alpha_0 = \alpha_1 \rightarrow \alpha_1 & \alpha_0 = \alpha_1 \rightarrow \alpha_1 \\
\alpha_1 = \alpha_3 \rightarrow \mathbb{B} & \alpha_1 = \mathbb{B} \rightarrow \mathbb{B} & \alpha_1 = \mathbb{B} \rightarrow \mathbb{B} & \Rightarrow \alpha_1 = \mathbb{B} \rightarrow \mathbb{B} \\
\Rightarrow \alpha_3 = \mathbb{B} & \Rightarrow \alpha_1 = \alpha_2 & \alpha_1 = \alpha_4 \rightarrow \alpha_5 & \alpha_1 = \alpha_4 \rightarrow \alpha_5 \\
\alpha_1 = \alpha_2 & \alpha_2 = \alpha_4 \rightarrow \alpha_5 & \alpha_1 = \alpha_6 \rightarrow \alpha_5 & \alpha_1 = \alpha_6 \rightarrow \alpha_5 \\
\alpha_2 = \alpha_4 \rightarrow \alpha_5 & \alpha_1 = \alpha_6 \rightarrow \alpha_5 & \alpha_1 = \alpha_7 \rightarrow \alpha_6 & \alpha_1 = \alpha_4 \rightarrow \alpha_6 \\
\alpha_1 = \alpha_6 \rightarrow \alpha_5 & \alpha_1 = \alpha_7 \rightarrow \alpha_6 & \Rightarrow \alpha_4 = \alpha_7 & \\
\alpha_1 = \alpha_7 \rightarrow \alpha_6 & \alpha_4 = \alpha_7 & & \\
\alpha_4 = \alpha_7 & & &
\end{array}$$

$$\begin{array}{lll}
\Rightarrow \alpha_0 = (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B} & \frac{\alpha_0 \mapsto (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}}{\Rightarrow \mathbb{B} \rightarrow \mathbb{B} = \alpha_4 \rightarrow \alpha_5} & \frac{\alpha_0 \mapsto (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}}{\Rightarrow \mathbb{B} = \alpha_4} \\
\mathbb{B} \rightarrow \mathbb{B} = \alpha_4 \rightarrow \alpha_5 & \mathbb{B} \rightarrow \mathbb{B} = \alpha_6 \rightarrow \alpha_5 & \mathbb{B} = \alpha_5 \\
\mathbb{B} \rightarrow \mathbb{B} = \alpha_6 \rightarrow \alpha_5 & \mathbb{B} \rightarrow \mathbb{B} = \alpha_4 \rightarrow \alpha_6 & \mathbb{B} \rightarrow \mathbb{B} = \alpha_6 \rightarrow \alpha_5 \\
\mathbb{B} \rightarrow \mathbb{B} = \alpha_4 \rightarrow \alpha_6 & & \mathbb{B} \rightarrow \mathbb{B} = \alpha_4 \rightarrow \alpha_6
\end{array}$$

$$\begin{array}{lll}
\frac{\alpha_0 \mapsto (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}}{\Rightarrow \mathbb{B} = \alpha_5} & \frac{\alpha_0 \mapsto (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}}{\Rightarrow \mathbb{B} \rightarrow \mathbb{B} = \alpha_6 \rightarrow \mathbb{B}} & \frac{\alpha_0 \mapsto (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}}{\mathbb{B} = \alpha_6} \\
\mathbb{B} \rightarrow \mathbb{B} = \alpha_6 \rightarrow \alpha_5 & \mathbb{B} \rightarrow \mathbb{B} = \mathbb{B} \rightarrow \alpha_6 & \Rightarrow \mathbb{B} = \mathbb{B} \\
\mathbb{B} \rightarrow \mathbb{B} = \mathbb{B} \rightarrow \alpha_6 & & \mathbb{B} \rightarrow \mathbb{B} = \mathbb{B} \rightarrow \alpha_6
\end{array}$$

$$\begin{array}{lll}
\frac{\alpha_0 \mapsto (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}}{\Rightarrow \mathbb{B} = \alpha_6} & \frac{\alpha_0 \mapsto (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}}{\Rightarrow \mathbb{B} \rightarrow \mathbb{B} = \mathbb{B} \rightarrow \mathbb{B}} & \frac{\alpha_0 \mapsto (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}}{} \\
\mathbb{B} \rightarrow \mathbb{B} = \mathbb{B} \rightarrow \alpha_6 & &
\end{array}$$

Hence the principal type is $(\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}$. Even though we find this expression for α_0 relatively early on, it was important to continue eliminating the other constraints in case they turned out to be inconsistent.

(d) $\lambda x. \lambda f. \text{if } f x \text{ then } f \text{ else } f$

$$\frac{\frac{\frac{x : \alpha_1, f : \alpha_3 \vdash f : \alpha_5 \rightarrow \mathbb{B}}{\text{VAR } [3]} \quad \frac{x : \alpha_1, f : \alpha_3 \vdash x : \alpha_5}{\text{VAR } [4]}}{x : \alpha_1, f : \alpha_3 \vdash f x : \mathbb{B}} \quad \frac{\frac{\frac{x : \alpha_1, f : \alpha_3 \vdash f : \alpha_4}{\text{VAR } [5]} \quad \frac{x : \alpha_1, f : \alpha_3 \vdash f : \alpha_4}{\text{VAR } [6]}}{x : \alpha_1, f : \alpha_3 \vdash \text{if } f x \text{ then } f \text{ else } f : \alpha_4}}{x : \alpha_1 \vdash \lambda f. \text{if } f x \text{ then } f \text{ else } f : \alpha_2} \quad \frac{}{\text{ABS } [1]} \quad \frac{}{\text{ABS } [2]}$$

where the following constraints are generated:

$$\begin{array}{l}
[1] : \alpha_0 = \alpha_1 \rightarrow \alpha_2 \\
[2] : \alpha_2 = \alpha_3 \rightarrow \alpha_4 \\
[3] : \alpha_3 = \alpha_5 \rightarrow \mathbb{B} \\
[4] : \alpha_1 = \alpha_5 \\
[5] : \alpha_3 = \alpha_4 \\
[6] : \alpha_3 = \alpha_4
\end{array}$$

(Recall that constraints form a set, so we can discard duplicated constraints).

We unify, this time recording the full substitution:

$$\begin{array}{c}
\begin{array}{c}
\text{⊢} \alpha_0 = \alpha_1 \rightarrow \alpha_2 \\
\alpha_2 = \alpha_3 \rightarrow \alpha_4 \\
\alpha_3 = \alpha_5 \rightarrow \mathbb{B} \\
\alpha_1 = \alpha_5 \\
\alpha_3 = \alpha_4
\end{array}
\quad
\frac{\alpha_0 \mapsto \alpha_1 \rightarrow \alpha_2}{\text{⊢} \alpha_2 = \alpha_3 \rightarrow \alpha_4}
\quad
\frac{\alpha_0 \mapsto \alpha_1 \rightarrow \alpha_3 \rightarrow \alpha_4 \quad \alpha_2 \mapsto \alpha_3 \rightarrow \alpha_4}{\text{⊢} \alpha_3 = \alpha_5 \rightarrow \mathbb{B}}
\quad
\frac{\alpha_0 \mapsto \alpha_1 \rightarrow (\alpha_5 \rightarrow \mathbb{B}) \rightarrow \alpha_4 \quad \alpha_2 \mapsto (\alpha_5 \rightarrow \mathbb{B}) \rightarrow \alpha_4 \quad \alpha_3 \mapsto \alpha_5 \rightarrow \mathbb{B}}{\text{⊢} \alpha_1 = \alpha_5}
\end{array}$$

$$\begin{array}{c}
\frac{\alpha_0 \mapsto \alpha_5 \rightarrow (\alpha_5 \rightarrow \mathbb{B}) \rightarrow \alpha_4 \quad \alpha_2 \mapsto (\alpha_5 \rightarrow \mathbb{B}) \rightarrow \alpha_4 \quad \alpha_3 \mapsto \alpha_5 \rightarrow \mathbb{B} \quad \alpha_1 \mapsto \alpha_5}{\text{⊢} \alpha_5 \rightarrow \mathbb{B} = \alpha_4}
\quad
\frac{\alpha_0 \mapsto \alpha_5 \rightarrow (\alpha_5 \rightarrow \mathbb{B}) \rightarrow \alpha_5 \rightarrow \mathbb{B} \quad \alpha_2 \mapsto (\alpha_5 \rightarrow \mathbb{B}) \rightarrow \alpha_4 \quad \alpha_3 \mapsto \alpha_5 \rightarrow \mathbb{B} \quad \alpha_1 \mapsto \alpha_5}{\alpha_4 \mapsto \alpha_5 \rightarrow \mathbb{B}}
\end{array}$$

Hence, the principal type is $\alpha \rightarrow (\alpha \rightarrow \mathbb{B}) \rightarrow \alpha \rightarrow \mathbb{B}$.

Question 3: Type-checking let-polymorphism

Give derivations to show that each of the following is a valid typing judgement (Curry-style λ -calculus with let-polymorphism).

(a) $x : \forall \alpha. \alpha \vdash x : \mathbb{N}$

$$\frac{}{x : \forall \alpha. \alpha \vdash x : \mathbb{N}} \text{VAR}$$

(b) $x : \forall \alpha, \beta. \alpha \vdash x : \mathbb{N}$

$$\frac{}{x : \forall \alpha, \beta. \alpha \vdash x : \mathbb{N}} \text{VAR}$$

since $\mathbb{N} = \alpha[\alpha \setminus \mathbb{N}, \beta \setminus \mathbb{N}]$.

(c) $f : \forall \alpha. \alpha \rightarrow \alpha \vdash \text{if } f \text{ true then } f \text{ else } \lambda x. x : \mathbb{N} \rightarrow \mathbb{N}$

$$\frac{\frac{\frac{}{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f : \mathbb{B} \rightarrow \mathbb{B}} \text{VAR} \quad \frac{}{f : \forall \alpha. \alpha \rightarrow \alpha \vdash \text{true} : \mathbb{B}} \text{VAR}}{\frac{}{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f \text{ true} : \mathbb{B}} \text{APP}} \quad \frac{}{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f : \mathbb{N} \rightarrow \mathbb{N}} \text{VAR}}{\frac{}{f : \forall \alpha. \alpha \rightarrow \alpha \vdash \text{if } f \text{ true then } f \text{ else } \lambda x. x : \mathbb{N} \rightarrow \mathbb{N}} \Pi_1} \text{ITE}$$

where Π_1 is

$$\frac{\frac{}{f : \forall \alpha. \alpha \rightarrow \alpha, x : \mathbb{N} \vdash x : \mathbb{N}} \text{VAR}}{f : \forall \alpha. \alpha \rightarrow \alpha \vdash \lambda x. x : \mathbb{N} \rightarrow \mathbb{N}} \text{ABS}$$

(d) $f : \forall \alpha. \alpha \rightarrow \alpha \vdash f f : \beta \rightarrow \beta$

$$\frac{\frac{}{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f : (\beta \rightarrow \beta) \rightarrow \beta \rightarrow \beta} \text{VAR} \quad \frac{}{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f : \beta \rightarrow \beta} \text{VAR}}{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f f : \beta \rightarrow \beta} \text{APP}$$

(e) $e : \forall \alpha. \alpha \rightarrow \alpha \rightarrow \mathbb{B}, x : \beta, y : \beta \vdash \text{if } e x y \text{ then succ zero else zero}$

Write $\Gamma = e : \forall \alpha. \alpha \rightarrow \alpha \rightarrow \mathbb{B}, x : \beta, y : \beta$.

$$\frac{\Pi_1 \quad \frac{\Gamma \vdash \text{zero} : \mathbb{N}}{\Gamma \vdash \text{succ zero} : \mathbb{N}}^{\text{S}} \quad \frac{\Gamma \vdash \text{zero} : \mathbb{N}}{\Gamma \vdash \text{zero} : \mathbb{N}}^{\text{Z}}}{\Gamma \vdash \text{if } e x y \text{ then succ zero else zero} : \mathbb{N}}^{\text{ITE}}$$

where Π_1 is

$$\frac{\frac{\Gamma \vdash e : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}}{\Gamma \vdash e x : \mathbb{B} \rightarrow \mathbb{B}}^{\text{VAR}} \quad \frac{\Gamma \vdash x : \mathbb{B}}{\Gamma \vdash e x : \mathbb{B}}^{\text{APP}} \quad \frac{\Gamma \vdash y : \mathbb{B}}{\Gamma \vdash e x y : \mathbb{B}}^{\text{APP}}}{\Gamma \vdash e x y : \mathbb{B}}^{\text{APP}}$$

(f) $\vdash \text{let } f = \lambda x. \lambda y. x \text{ in if } f \text{ true false then } f \text{ zero true else } f \text{ zero } f : \mathbb{N}$

Let's write $\Gamma = f : \forall \alpha, \beta. \alpha \rightarrow \beta \rightarrow \alpha$ to save space.

$$\frac{\frac{\frac{x : \alpha, y : \beta \vdash x : \alpha}{x : \alpha \vdash \lambda y. x : \beta \rightarrow \alpha}^{\text{VAR}}}{x : \alpha \vdash \lambda y. x : \beta \rightarrow \alpha}^{\text{ABS}} \quad \frac{\frac{\Gamma \vdash f : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}}{\Gamma \vdash f \text{ true} : \mathbb{B}}^{\text{VAR}} \quad \frac{\Gamma \vdash \text{true} : \mathbb{B}}{\Gamma \vdash f \text{ true} : \mathbb{B}}^{\text{APP}} \quad \frac{\Gamma \vdash \text{false} : \mathbb{B}}{\Gamma \vdash f \text{ false} : \mathbb{B}}^{\text{F}}}{\Gamma \vdash f \text{ true false} : \mathbb{B}}^{\text{APP}} \quad \frac{\Pi_1 \quad \Pi_2}{\Gamma \vdash \text{if } f \text{ true false then } f \text{ zero true else } f \text{ zero } f : \mathbb{N}}^{\text{ITE}}}{\vdash \text{let } f = \lambda x. \lambda y. x \text{ in if } f \text{ true false then } f \text{ zero true else } f \text{ zero } f : \mathbb{N}}^{\text{LET}}$$

where Π_1 is

$$\frac{\frac{\Gamma \vdash f : \mathbb{N} \rightarrow \mathbb{B} \rightarrow \mathbb{N}}{\Gamma \vdash f \text{ zero} : \mathbb{B} \rightarrow \mathbb{N}}^{\text{VAR}} \quad \frac{\Gamma \vdash \text{zero} : \mathbb{N}}{\Gamma \vdash f \text{ zero} : \mathbb{B} \rightarrow \mathbb{N}}^{\text{Z}}}{\Gamma \vdash f \text{ zero true} : \mathbb{N}}^{\text{APP}} \quad \frac{\Gamma \vdash \text{true} : \mathbb{B}}{\Gamma \vdash f \text{ zero true} : \mathbb{N}}^{\text{T}}$$

and Π_2 is

$$\frac{\frac{\Gamma \vdash f : \mathbb{N} \rightarrow \gamma \rightarrow \mathbb{N}}{\Gamma \vdash f \text{ zero} : \gamma \rightarrow \mathbb{N}}^{\text{VAR}} \quad \frac{\Gamma \vdash \text{zero} : \mathbb{N}}{\Gamma \vdash f \text{ zero} : \gamma \rightarrow \mathbb{N}}^{\text{Z}}}{\Gamma \vdash f \text{ zero } f : \mathbb{N}}^{\text{APP}} \quad \frac{\Gamma \vdash f : \gamma}{\Gamma \vdash f \text{ zero } f : \mathbb{N}}^{\text{VAR}}$$

where we could choose any other type instead of the type variable γ and still have a valid derivation.

Question 4: type-inference for let-polymorphism

For each of the following terms (Curry-style λ -calculus with let-polymorphism) find a type, type environment, and typing derivation, or else show that none exists. Try to find the most general type in each case.

(a) $x y$

$$\frac{\frac{x : \forall \alpha, \beta. \alpha \rightarrow \beta, y : \forall \alpha. \alpha \vdash x : \alpha \rightarrow \beta}{x : \forall \alpha, \beta. \alpha \rightarrow \beta, y : \forall \alpha. \alpha \vdash x y : \beta}^{\text{VAR}} \quad \frac{x : \forall \alpha, \beta. \alpha \rightarrow \beta, y : \forall \alpha. \alpha \vdash y : \alpha}{x : \forall \alpha, \beta. \alpha \rightarrow \beta, y : \forall \alpha. \alpha \vdash x y : \beta}^{\text{VAR}}}{x : \forall \alpha, \beta. \alpha \rightarrow \beta, y : \forall \alpha. \alpha \vdash x y : \beta}^{\text{APP}}$$

This is the most general type, but we could also use a more general typing environment, i.e. $x : \forall \alpha. \alpha, y : \forall \alpha. \alpha$.

(b) `if iszero  $x$  then  $f$   $x$  else  $f$  true`

Write $\Gamma = x : \sigma_0, f : \sigma_1$. Let us try to find a derivation tree, leaving some types as unknowns.

$$\frac{\frac{\overline{\Gamma \vdash x : \mathbb{N}}^{\text{VAR}}}{\Gamma \vdash \text{iszero } x : \mathbb{B}}^{\text{I}} \quad \frac{\frac{\overline{\Gamma \vdash f : \alpha_1 \rightarrow \alpha_0}^{\text{VAR}} \quad \overline{\Gamma \vdash x : \alpha_1}^{\text{VAR}}}{\Gamma \vdash f x : \alpha_0}^{\text{APP}} \quad \frac{\overline{\Gamma \vdash f : \mathbb{B} \rightarrow \alpha_0}^{\text{VAR}} \quad \overline{\Gamma \vdash \text{true} : \mathbb{B}}^{\text{T}}}{\Gamma \vdash f \text{true} : \alpha_0}^{\text{APP}}}{\Gamma \vdash \text{if iszero } x \text{ then } f x \text{ else } f \text{true} : \alpha_0}^{\text{ITE}}$$

It is not hard to find σ_0 and σ_1 which make this whole derivation valid already with α_0 as the final type. One possibility is with $\sigma_0 = \mathbb{N}$ and $\sigma_1 = \forall \alpha, \beta. \alpha \rightarrow \beta$. Another is $\sigma_0 = \forall \alpha. \alpha$ and $\sigma_1 = \forall \alpha. \mathbb{B} \rightarrow \alpha$. But also $\sigma_0 = \sigma_1 = \forall \alpha. \alpha$ works. This is not a nice solution, however. We rather hope that types like α and $\alpha \rightarrow \beta$ should not have any closed terms.

(c) `$\lambda f. \text{succ}(f x)$`

$$\frac{\frac{\frac{\overline{x : \sigma, f : \alpha_1 \vdash f : \alpha_2 \rightarrow \mathbb{N}}^{\text{VAR}} \quad \overline{x : \sigma, f : \alpha_1 \vdash x : \alpha_2}^{\text{VAR}}}{\overline{x : \sigma, f : \alpha_1 \vdash f x : \mathbb{N}}^{\text{APP}}}{\overline{x : \sigma, f : \alpha_1 \vdash \text{succ}(f x) : \mathbb{N}}^{\text{S}}}{\overline{x : \sigma \vdash \lambda f. \text{succ}(f x) : \alpha_0}^{\text{ABS}}}$$

where we need $\alpha_0 = \alpha_1 \rightarrow \mathbb{N}$. We can make this substitution, and also replace α_1 with $\alpha_2 \rightarrow \mathbb{N}$ as implied by one of the VAR rule instances.

$$\frac{\frac{\overline{x : \sigma, f : \alpha_2 \rightarrow \mathbb{N} \vdash f : \alpha_2 \rightarrow \mathbb{N}}^{\text{VAR}} \quad \overline{x : \sigma, f : \alpha_2 \rightarrow \mathbb{N} \vdash x : \alpha_2}^{\text{VAR}}}{\overline{x : \sigma, f : \alpha_2 \rightarrow \mathbb{N} \vdash f x : \mathbb{N}}^{\text{APP}}}{\overline{x : \sigma, f : \alpha_2 \rightarrow \mathbb{N} \vdash \text{succ}(f x) : \mathbb{N}}^{\text{S}}}{\overline{x : \sigma \vdash \lambda f. \text{succ}(f x) : (\alpha_2 \rightarrow \mathbb{N}) \rightarrow \mathbb{N}}^{\text{ABS}}}$$

The only requirement now is that σ needs to be able to specialize to α_2 . For example, if $x : T$ in the typing environment, then we can type the term with $(T \rightarrow \mathbb{N}) \rightarrow \mathbb{N}$.

(d) `let  $f = \lambda x. x x$  in  $f (\lambda y. y)$  true`

Let's try to write a derivation

$$\frac{\frac{\vdots}{\vdash \lambda x. x x : T} \quad \frac{\vdots}{f : \forall FV(T). T \vdash f (\lambda y. y) \text{true} : U}}{\vdash \text{let } f = \lambda x. x x \text{ in } f (\lambda y. y) \text{true} : U}^{\text{LET}}$$

But the left branch cannot be completed. We would need $T = T_1 \rightarrow T_2$, where

$$\frac{\overline{x : T_1 \vdash x : T_3 \rightarrow T_2}^{\text{VAR}} \quad \overline{x : T_1 \vdash x : T_3}^{\text{VAR}}}{\overline{x : T_1 \vdash x x : T_2}^{\text{APP}}}{\vdash \lambda x. x x : T_1 \rightarrow T_2}^{\text{ABS}}$$

for some T_3 but these constraints can't be solved.

(e) `let f = λx.x in f (λy.y) true`

For example, we can type $\lambda x.x$ with

$$\frac{\overline{x : \alpha \vdash x : \alpha} \text{VAR}}{\lambda x.x : \alpha \rightarrow \alpha} \text{ABS}$$

using a type variable α , and therefore we can try to begin a type derivation

$$\frac{\frac{\overline{x : \alpha \vdash x : \alpha} \text{VAR}}{\vdash \lambda x.x : \alpha \rightarrow \alpha} \text{ABS} \quad \frac{\vdots}{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f (\lambda y.y) \text{true} : \alpha_0} \text{APP}}{\vdash \text{let } f = \lambda x.x \text{ in } f (\lambda y.y) \text{true} : \alpha_0} \text{LET}$$

where α_0 is still unknown. We try to fill in the ellipsis:

$$\frac{\frac{\overline{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f : \alpha_1 \rightarrow \mathbb{B} \rightarrow \alpha_0} \text{VAR} \quad \frac{\overline{f : \forall \alpha. \alpha \rightarrow \alpha, y : \alpha_2 \vdash y : \alpha_2} \text{VAR}}{\overline{f : \forall \alpha. \alpha \rightarrow \alpha \vdash \lambda y.y : \alpha_1} \text{ABS}} \text{APP}}{\overline{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f (\lambda y.y) : \mathbb{B} \rightarrow \alpha_0} \text{APP}} \quad \frac{\overline{f : \forall \alpha. \alpha \rightarrow \alpha \vdash \text{true} : \mathbb{B}} \text{T}}{\overline{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f (\lambda y.y) \text{true} : \alpha_0} \text{APP}}$$

which requires the following constraints be satisfied:

$$\begin{aligned} \alpha_1 &= \alpha_2 \rightarrow \alpha_2 \\ \alpha_1 \rightarrow \mathbb{B} \rightarrow \alpha_0 &= (\alpha \rightarrow \alpha)[\alpha \backslash T] \end{aligned}$$

for some type T . We can eliminate α_1 , so we just need

$$(\alpha_2 \rightarrow \alpha_2) \rightarrow \mathbb{B} \rightarrow \alpha_0 = T \rightarrow T$$

for some T . We can turn this into a pair of constraints

$$\begin{aligned} \alpha_2 \rightarrow \alpha_2 &= T \\ \mathbb{B} \rightarrow \alpha_0 &= T \end{aligned}$$

from which we read off $\alpha_2 = \mathbb{B} = \alpha_0$. Indeed this does give a valid type derivation, explicitly:

$$\frac{\frac{\overline{x : \alpha \vdash x : \alpha} \text{VAR}}{\vdash \lambda x.x : \alpha \rightarrow \alpha} \text{ABS} \quad \Pi_1}{\vdash \text{let } f = \lambda x.x \text{ in } f (\lambda y.y) \text{true} : \mathbb{B}} \text{LET}$$

where Π_1 is:

$$\frac{\frac{\overline{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}} \text{VAR} \quad \frac{\overline{f : \forall \alpha. \alpha \rightarrow \alpha, y : \mathbb{B} \vdash y : \mathbb{B}} \text{VAR}}{\overline{f : \forall \alpha. \alpha \rightarrow \alpha \vdash \lambda y.y : \mathbb{B} \rightarrow \mathbb{B}} \text{ABS}} \text{APP}}{\overline{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f (\lambda y.y) : \mathbb{B} \rightarrow \mathbb{B}} \text{APP}} \quad \frac{\overline{f : \forall \alpha. \alpha \rightarrow \alpha \vdash \text{true} : \mathbb{B}} \text{T}}{\overline{f : \forall \alpha. \alpha \rightarrow \alpha \vdash f (\lambda y.y) \text{true} : \mathbb{B}} \text{APP}}$$

Question 5: Type-checking System F terms

Given derivations to show that each of the following is a valid typing judgement (Church-style System F terms).

(a) $x : \mathbb{N} \vdash (\lambda f : \forall \alpha. \alpha \rightarrow \alpha. f\{\mathbb{N}\} x) (\lambda \alpha. \lambda y : \alpha. y) : \mathbb{N}$

$$\frac{\frac{\frac{\overline{x : \mathbb{N}, f : \forall \alpha. \alpha \rightarrow \alpha \vdash f : \forall \alpha. \alpha \rightarrow \alpha} \text{VAR}}{x : \mathbb{N}, f : \forall \alpha. \alpha \rightarrow \alpha \vdash f\{\mathbb{N}\} : \mathbb{N} \rightarrow \mathbb{N}} \text{TAPP} \quad \frac{\overline{x : \mathbb{N}, f : \forall \alpha. \alpha \rightarrow \alpha \vdash x : \mathbb{N}} \text{VAR}}{\frac{x : \mathbb{N}, f : \forall \alpha. \alpha \rightarrow \alpha \vdash f\{\mathbb{N}\} x : \mathbb{N}} \text{APP}} \text{ABS} \quad \frac{\frac{\overline{x : \mathbb{N}, y : \alpha \vdash y : \alpha} \text{VAR}}{x : \mathbb{N} \vdash \lambda y : \alpha. y : \alpha \rightarrow \alpha} \text{ABS}}{x : \mathbb{N} \vdash \lambda \alpha. \lambda y : \alpha. y : \forall \alpha. \alpha \rightarrow \alpha} \text{TABS}}{x : \mathbb{N} \vdash (\lambda f : \forall \alpha. \alpha \rightarrow \alpha. f\{\mathbb{N}\} x) (\lambda \alpha. \lambda y : \alpha. y) : \mathbb{N}} \text{APP}$$

(b) $\vdash \lambda \alpha. \lambda x : \alpha. \lambda y : \alpha. x : \forall \alpha. \alpha \rightarrow \alpha \rightarrow \alpha$

$$\frac{\frac{\frac{\overline{x : \alpha, y : \alpha \vdash x : \alpha} \text{VAR}}{x : \alpha \vdash \lambda y : \alpha. x : \alpha \rightarrow \alpha} \text{ABS}}{\vdash \lambda x : \alpha. \lambda y : \alpha. x : \alpha \rightarrow \alpha \rightarrow \alpha} \text{ABS}}{\vdash \lambda \alpha. \lambda x : \alpha. \lambda y : \alpha. x : \forall \alpha. \alpha \rightarrow \alpha \rightarrow \alpha} \text{TABS}$$

(c) $x : \alpha \vdash \lambda \gamma. \lambda f : \alpha \rightarrow \gamma. \lambda g : \beta \rightarrow \gamma. f x : \forall \gamma. (\alpha \rightarrow \gamma) \rightarrow (\beta \rightarrow \gamma) \rightarrow \gamma$

$$\frac{\frac{\frac{\overline{x : \alpha, f : \alpha \rightarrow \gamma, g : \beta \rightarrow \gamma \vdash f : \alpha \rightarrow \gamma} \text{VAR}}{x : \alpha, f : \alpha \rightarrow \gamma, g : \beta \rightarrow \gamma \vdash f x : \gamma} \text{APP}}{\frac{x : \alpha, f : \alpha \rightarrow \gamma, g : \beta \rightarrow \gamma \vdash f x : \gamma} \text{ABS}} \text{ABS} \quad \frac{\frac{x : \alpha, f : \alpha \rightarrow \gamma \vdash \lambda g : \beta \rightarrow \gamma. f x : (\beta \rightarrow \gamma) \rightarrow \gamma} \text{ABS}}{x : \alpha \vdash \lambda f : \alpha \rightarrow \gamma. \lambda g : \beta \rightarrow \gamma. f x : (\alpha \rightarrow \gamma) \rightarrow (\beta \rightarrow \gamma) \rightarrow \gamma} \text{TABS}}{x : \alpha \vdash \lambda \gamma. \lambda f : \alpha \rightarrow \gamma. \lambda g : \beta \rightarrow \gamma. f x : \forall \gamma. (\alpha \rightarrow \gamma) \rightarrow (\beta \rightarrow \gamma) \rightarrow \gamma} \text{TABS}$$

(d) $\vdash \lambda f : \forall \alpha. \alpha. f\{\beta\} : (\forall \alpha. \alpha) \rightarrow \beta$

$$\frac{\frac{\frac{\overline{f : \forall \alpha. \alpha \vdash f : \forall \alpha. \alpha} \text{VAR}}{f : \forall \alpha. \alpha \vdash f\{\beta\} : \beta} \text{TAPP}}{\vdash \lambda f : \forall \alpha. \alpha. f\{\beta\} : (\forall \alpha. \alpha) \rightarrow \beta} \text{ABS}$$

Question 6: Small-step CBV semantics of System F terms

Reduce each of the following (Church-style, System F) terms to a CBV normal form.

(a) $(\lambda f : \forall \alpha. \alpha \rightarrow \alpha. f\{\beta\}) (\lambda \alpha. \lambda x : \alpha. x)$

$$\begin{aligned} & (\lambda f : \forall \alpha. \alpha \rightarrow \alpha. f\{\beta\}) (\lambda \alpha. \lambda x : \alpha. x) \\ & \rightarrow_v (\lambda \alpha. \lambda x : \alpha. x) \{\beta\} \\ & \rightarrow_v \lambda x : \beta. x \end{aligned}$$

(b) $(\lambda g : \forall \gamma. (\mathbb{N} \rightarrow \mathbb{B} \rightarrow \gamma) \rightarrow \gamma. g\{\mathbb{N}\} \lambda x : \mathbb{N}. \lambda y : \mathbb{B}. x) (\lambda \gamma. \lambda f : (\mathbb{N} \rightarrow \mathbb{B} \rightarrow \gamma) \rightarrow \gamma. f \text{ zero true})$

$(\lambda g : \forall \gamma. (\mathbb{N} \rightarrow \mathbb{B} \rightarrow \gamma) \rightarrow \gamma. g\{\mathbb{N}\} \lambda x : \mathbb{N}. \lambda y : \mathbb{B}. x) (\lambda \gamma. \lambda f : (\mathbb{N} \rightarrow \mathbb{B} \rightarrow \gamma) \rightarrow \gamma. f \text{ zero true})$
 $\rightarrow_v (\lambda \gamma. \lambda f : (\mathbb{N} \rightarrow \mathbb{B} \rightarrow \gamma) \rightarrow \gamma. f \text{ zero true}) \{\mathbb{N}\} \lambda x : \mathbb{N}. \lambda y : \mathbb{B}. x$
 $\rightarrow_v (\lambda f : (\mathbb{N} \rightarrow \mathbb{B} \rightarrow \mathbb{N}) \rightarrow \mathbb{N}. f \text{ zero true}) \lambda x : \mathbb{N}. \lambda y : \mathbb{B}. x$
 $\rightarrow_v (\lambda x : \mathbb{N}. \lambda y : \mathbb{B}. x) \text{ zero true}$
 $\rightarrow_v (\lambda y : \mathbb{B}. \text{zero}) \text{ true}$
 $\rightarrow_v \text{ zero}$